

RPCA-Enhanced Anomaly Detection on the UCSD Pedestrian Dataset

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Abstract

This project demonstrates how two classical matrix-decomposition techniques from data mining — Robust Principal Component Analysis (RPCA) and the Singular Value Decomposition (SVD) — can be used to enhance and analyse a deep-learning anomaly detection system. We investigate whether RPCA, applied as a per-clip preprocessing step, improves frame-level anomaly detection on the UCSD Pedestrian benchmark. A UNet future-frame predictor is trained under four input/target configurations on Ped1 and Ped2. The headline result is that RPCA preprocessing yields a substantial AUC gain on Ped2 ($0.582 \rightarrow 0.886$) and a smaller gain on Ped1 ($0.702 \rightarrow 0.783$). A non-learned baseline using only the per-frame ℓ_1 -norm of RPCA’s sparse component achieves 0.812 (Ped2) and 0.718 (Ped1) — capturing the majority of the benefit attributable to the data representation, with the learned predictor adding roughly 7 percentage points on top. An input-side / target-side ablation (`raw_to_S`) shows that the predictor’s contribution is realised only when S is provided as input, not as target. A naïve SVD weight-compression study on the trained predictor produces a non-monotone AUC curve, confirming that finetuning is required after rank truncation if model compression is the goal.

1 Background and Motivation

Video anomaly detection identifies frames or regions in surveillance footage that deviate from learned patterns of normal activity. The UCSD Pedestrian Anomaly Dataset [4] is one of the most widely used benchmarks for this task. It consists of two subsets — **Ped1** and **Ped2** — captured by stationary cameras overlooking pedestrian walkways. In normal frames only pedestrians appear; anomalies are non-pedestrian entities (cyclists, skateboarders, small carts, vehicles) or unusual pedestrian motion patterns.

The two subsets differ in ways that matter for our analysis:

Property	Ped1	Ped2
Resolution	158×238	240×360
Train clips	34	16
Test clips	36	12
Camera angle	Walkway approaches with foreshortening	Walkway from the side, near-orthogonal
Crowd density	Dense — pedestrians fill most frames	Sparse — few or no pedestrians per frame
Anomaly types	Cyclists, small carts	Cyclists, skateboarders, occasional vehicles

Table 1: Differences between the two UCSD subsets.

Frame-level and pixel-level ground-truth annotations are provided for the test set; we use frame-level only.

The fundamental structure of this anomaly-detection problem is the separation of *rare, transient deviations* from a *persistent, structured background*. This is the precise mathematical structure that Robust PCA recovers — which makes UCSD an unusually clean test case for RPCA-based methods. The next section explains why. More broadly, this project is a case study in how the classical matrix-decomposition tools covered in the course — RPCA and SVD — can be applied both to enhance a modern deep-learning pipeline (as preprocessing) and to analyse it (as a model-compression diagnostic), with the empirical study itself revealing where the contribution of each component actually lies.

2 Robust PCA: What It Is and Why It Fits This Problem

2.1 From Classical PCA to Robust PCA

Classical Principal Component Analysis approximates a data matrix X by its best rank- k approximation L , computed from the truncated SVD. If we write $X = U\Sigma V^\top$ with singular values $\sigma_1 \geq \sigma_2 \geq \dots$, then keeping only the top k components gives the closest rank- k matrix to X in Frobenius norm. PCA is the workhorse of dimensionality reduction precisely because real data often has a low-rank structure: most of the variance is concentrated in a few directions.

But classical PCA has a known weakness: it is *not* robust to outliers. A single corrupted entry can pull the top singular vectors arbitrarily far from where they would be on clean data. For applications where the data contains a small number of dramatic deviations from an otherwise low-rank structure, classical PCA gives the wrong answer.

Robust PCA, formalized by Candès, Li, Ma, and Wright [1], addresses this by changing the model. Instead of approximating X as a single low-rank matrix, RPCA assumes that X is the sum of two parts:

$$X = L + S \tag{1}$$

where L is *exactly* low-rank and S is *sparse* — most of its entries are zero, but the non-zero entries can be arbitrarily large. The two components are recovered by solving the convex optimization problem

$$\min_{L,S} \|L\|_* + \lambda \|S\|_1 \quad \text{subject to} \quad X = L + S \tag{2}$$

The nuclear norm $\|L\|_*$ (the sum of singular values of L) is a convex relaxation of rank — minimizing it pushes L toward low rank. The ℓ_1 norm $\|S\|_1$ (the sum of absolute values of S 's entries) is a convex relaxation of sparsity — minimizing it pushes S toward having few non-zero entries. The parameter λ balances the two; [1] show that the choice $\lambda = 1/\sqrt{\max(n_1, n_2)}$ has strong recovery guarantees and is parameter-free in practice.

Remarkably, under mild conditions, RPCA recovers L and S *exactly* — not approximately — even when S is large in magnitude, as long as it is sparse enough [1]. The decomposition is unique and identifiable.

2.2 Why Surveillance Video Is Tailor-Made for RPCA

The original RPCA paper used surveillance video as one of its motivating examples [1]. The intuition is direct: imagine vectorizing each frame of a video clip into a column and stacking those columns into a matrix X . Each column is one frame; each row tracks one pixel over time.

If the camera is stationary and the scene has a stable background:

- The background is the *same* in every column. Columns are highly correlated — the matrix has a strong low-rank structure. This is the L component.
- Moving objects appear in only a small subset of pixels at any moment, and they move from frame to frame. Their contribution is sparse in the pixel domain — a few bright spots against a mostly-zero canvas. This is the S component.

The RPCA decomposition therefore separates the video into “what stays the same” (background, L) and “what changes transiently” (moving content, S). For the UCSD anomaly detection task, this is essentially free foreground extraction without any motion model, optical flow, or learned segmentation.

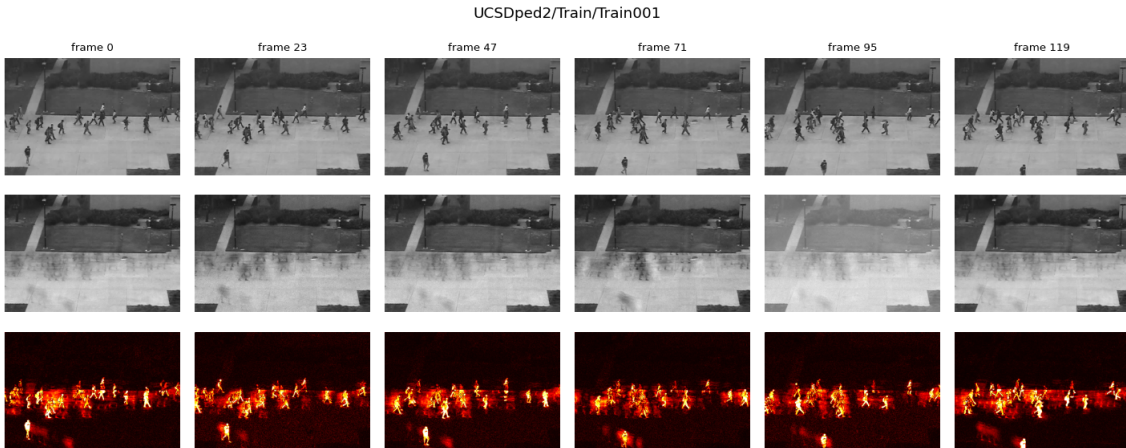


Figure 1: RPCA decomposition of a Ped2 training clip (Train001), shown across six frames. Top row: raw frames X . Middle row: recovered low-rank background L . Bottom row: sparse foreground $|S|$. The pedestrian crowd is cleanly isolated in $|S|$ across all frames while L remains a clean empty walkway.

2.3 Why the Singular Spectrum of UCSD Tells Us RPCA Will Work

Before training any model, we can verify that the RPCA assumption holds on the actual data by inspecting the singular spectrum of one clip. If X is well-approximated by a low-rank matrix plus a sparse component, then the singular values of X should drop off sharply after the first few — most of the energy lives in a small number of directions, with the rest representing the sparse foreground.

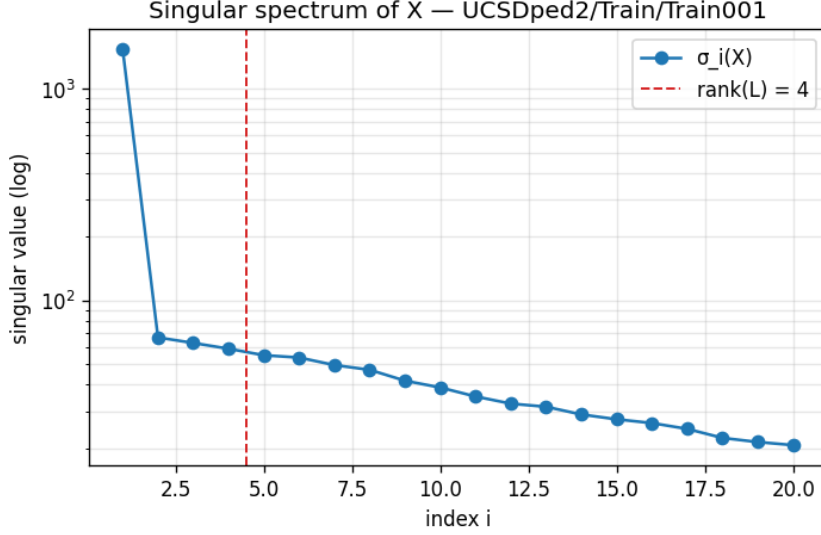


Figure 2: Singular spectrum $\sigma_i(X)$ of UCSDped2/Train001, with the rank of L recovered by RPCA (rank 4) marked. The first singular value (~ 1500) is roughly $20\times$ larger than the second, confirming the empty walkway is essentially a rank-1 image with small departures from it.

This is a strong empirical justification for using RPCA on UCSD: the data exhibits exactly the structure RPCA assumes.

2.4 How RPCA Is Solved in Practice

RPCA’s optimization problem has no closed form, but it can be solved efficiently with the inexact Augmented Lagrange Multiplier (ALM) method of [3]. The algorithm alternates between two updates:

- **Update L** by applying singular value thresholding (SVT) — compute the SVD, shrink each singular value toward zero by $1/\mu$, and discard those that fall to zero. This is where SVD enters the algorithm directly.
- **Update S** by applying soft thresholding to the residual $X - L$, shrinking each entry toward zero by λ/μ .

After a small number of iterations the residual $\|X - L - S\|_F$ falls below a tolerance and the algorithm terminates. Each iteration costs one full SVD of an $H \cdot W \times T$ matrix, which dominates the runtime.

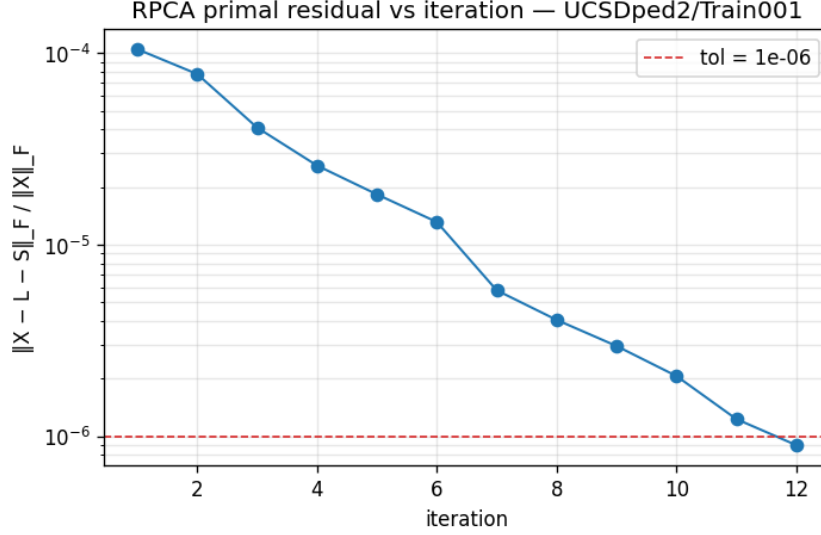


Figure 3: RPCA primal residual $\|X - L - S\|_F / \|X\|_F$ versus iteration on UCSDped2/Train001 (log y-axis). Convergence is monotone and reaches tolerance 10^{-6} in 12 iterations.

2.5 Connection to Course Material

This project draws on three threads of the course:

- **SVD** is the operational core of the SVT step — every RPCA iteration is built on a singular value decomposition, and the algorithm’s behavior is determined by which singular directions survive the soft-shrinkage. SVD reappears in §5.6 as a model-compression tool applied to the trained network’s weight matrices.
- **RPCA** is the central preprocessing operation, and the project treats its empirical behavior on a real benchmark as the primary contribution.
- **Compressed-sensing intuition** motivates the project at the conceptual level: the assumption that anomalous content is sparse in the pixel domain (after background removal) parallels the sparsity-in-a-basis assumption underlying compressed sensing reconstruction.

3 Research Questions

RQ1. Does RPCA-based foreground extraction, applied as preprocessing, improve frame-level AUC of a UNet future-frame predictor on UCSD Ped1 and Ped2 versus training the same architecture on raw frames?

RQ2. Which RPCA component carries the most discriminative signal — sparse foreground S , low-rank background L , or their concatenation $[L, S]$?

RQ3. How do the gains differ between Ped1 (low resolution, dense crowds) and Ped2 (higher resolution, sparse scenes)?

RQ4. Can SVD-based low-rank approximation of the trained predictor’s weights reduce model size with limited AUC degradation, providing a unified low-rank perspective spanning preprocessing, classical decomposition, and model compression?

A diagnostic question, added after the main experiments were complete, is **RQ-D**: How much of the AUC gain attributable to RPCA preprocessing comes from the representation itself (i.e., is achievable without any learned model), and how much comes from the predictor learning

useful structure on top? This question turns out to be central to the interpretation of all other results.

4 Methodology

4.1 Data Pipeline

We use the UCSD Pedestrian Anomaly Dataset as distributed on Kaggle ([orville/ucsd-anomaly-dataset](#)). For each video clip, frames are loaded at native resolution and converted to grayscale (UCSD is provided in grayscale). We evaluate at the frame level using the provided ground-truth annotations; pixel-level evaluation is not pursued.

One frame in UCSDped1/Test017 (142.tif) is corrupted in the distributed archive; we substitute the nearest valid frame (140) and log the substitution.

4.2 RPCA Preprocessing in Detail

For each clip, frames are vectorized and stacked column-wise into $X \in \mathbb{R}^{(H \cdot W) \times T}$, where T is the number of frames. We apply the inexact ALM solver [3] with $\mu_0 = n_1 n_2 / (4 \|X\|_1)$ as the initial penalty parameter and $\lambda = 1 / \sqrt{\max(n_1, n_2)}$ as the universal sparsity weight from [1]. RPCA is applied **per clip** rather than globally: each clip has its own static background, and a global decomposition would lose this per-scene structure.

Across all four splits (Ped1/Ped2 $\times$ Train/Test), the solver converged in an average of 8.7–10.8 iterations, with worst-case primal residual below 10^{-6} and per-clip wall time of 4–14 seconds. The recovered rank of L is small — typically 4 — confirming that the empty-walkway background is, in practice, very nearly rank 1 with three additional directions absorbing slow lighting variation and camera noise.

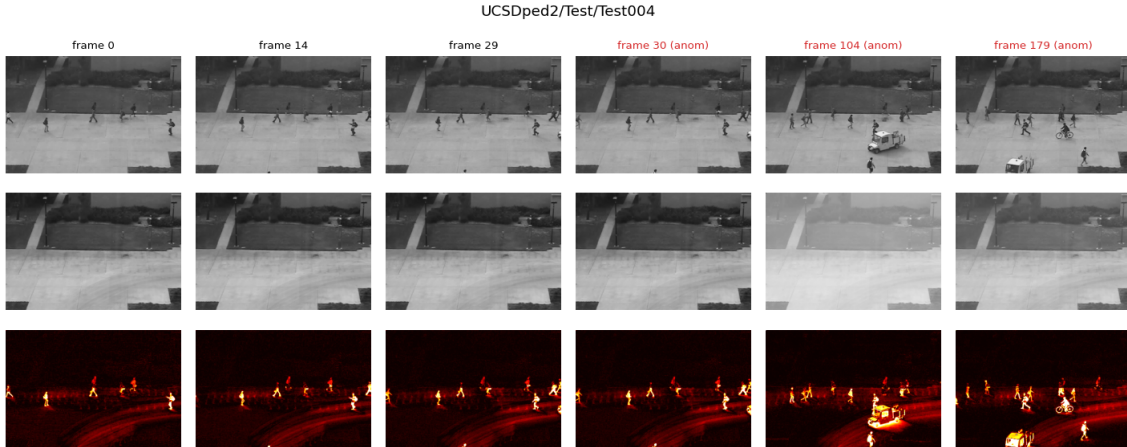


Figure 4: Qualitative L/S decomposition across six frames of UCSDped2/Test/Test004, three normal and three anomalous (anomalous frame titles in red). The top row shows X , the middle row shows L , the bottom row shows $|S|$. Note that L is essentially identical across all six frames — including those containing a vehicle (frames 30, 104) and a cyclist (frame 179) — while $|S|$ cleanly highlights the moving content. This is the most direct visual confirmation that RPCA’s modeling assumption holds on UCSD.

4.3 Phase 1 (Baseline) and Phase 2 (Predictor)

A per-frame convolutional autoencoder on raw frames, evaluated as an early baseline, achieved approximately 0.67 AUC on Ped2. Per-frame autoencoding does not provide a setting where RPCA is unambiguously useful, since both raw and S-domain reconstruction can be learned to similar quality on normal data. We therefore moved to **future-frame prediction**: the model receives a window of past frames and predicts the next frame; anomaly score is the per-frame reconstruction error against the actual next frame. Prediction is a task where motion — exactly what RPCA isolates — is what must be predicted.

4.4 Model and Training

We use a UNet with `base_channels = 32` (4.49M parameters), input window of 4 frames, single-channel grayscale output. Loss is the sum of an L2 reconstruction term and a gradient term ($\lambda_{\text{grad}} = 1.0$) penalizing differences in spatial gradients between prediction and target. Optimizer is Adam at learning rate 2×10^{-4} , with 3 epochs of linear warmup followed by cosine decay. Batch size 16 (Ped2) or 32 (Ped1), training for up to 100 epochs with early stopping (patience 10). Validation split is held out at the clip level (15% Ped2, 10% Ped1) so that no clip is shared between training and validation.

The same architecture, hyperparameters, and seed are used across all four input/target configurations. Only the input and target tensors change.

4.5 The Four Variants

Variant	Input	Target	RPCA acts on
raw	Raw frames	Raw frame	Nothing (baseline)
S	Sparse component	Sparse component	Both sides
LS	$[L, S]$ concatenated	Raw frame	Input only (richer representation)
raw_to_S	Raw frames	Sparse component	Target only

Table 2: The four input/target configurations. `raw_to_S` is the critical ablation: it isolates whether the gain comes from cleaner *input* or from a cleaner *target* domain.

4.6 Evaluation Protocol

We report frame-level AUC (global), per-clip AUC (mean of per-clip AUCs), and EER. Per-clip AUC on Ped2 is computed over 8 of 12 test clips: clips Test008–Test011 contain only anomalous frames (100% anomaly fraction by GT) so per-clip ROC is undefined.

4.7 The $\|S\|_1$ Diagnostic Baseline

Because of the way the Phase 2 results came out (§5.2), we added one further measurement: the per-frame ℓ_1 norm of the cached S component, $\|S_t\|_1$, used directly as an anomaly score with no model. This produces an upper bound on what is achievable from the RPCA representation alone — any predictor’s gain over this baseline is attributable to learning, not to the representation.

5 Results

5.1 Sanity Check: RPCA Decomposition Holds at Anomalous Frames

Figure 4 (above) already shows the key qualitative finding: L is essentially constant across all six frames spanning normal and anomalous content, while $|S|$ tracks moving content faithfully. The vehicle in frames 30 and 104, and the cyclist in frame 179, are clearly visible in the sparse component — confirming that anomalous moving objects are recovered as part of S rather than corrupting L .

5.2 Frame-Level AUC

Dataset	Variant	AUC (global)	AUC (per-clip)	EER
Ped2	raw	0.582	0.685	0.454
Ped2	S	0.886	0.942	0.164
Ped2	LS	0.418	0.490	0.583
Ped2	raw_to_S	0.546	0.546	0.498
Ped2	$\ S\ _1$ (no model)	0.812	0.813	—
Ped1	raw	0.702	0.771	0.350
Ped1	S	0.776	0.814	0.277
Ped1	LS	0.783	0.824	0.266
Ped1	raw_to_S	0.532	0.594	0.486
Ped1	$\ S\ _1$ (no model)	0.718	0.726	—

Table 3: Frame-level AUC, per-clip AUC, and EER across all variants and the non-learned $\|S\|_1$ baseline.

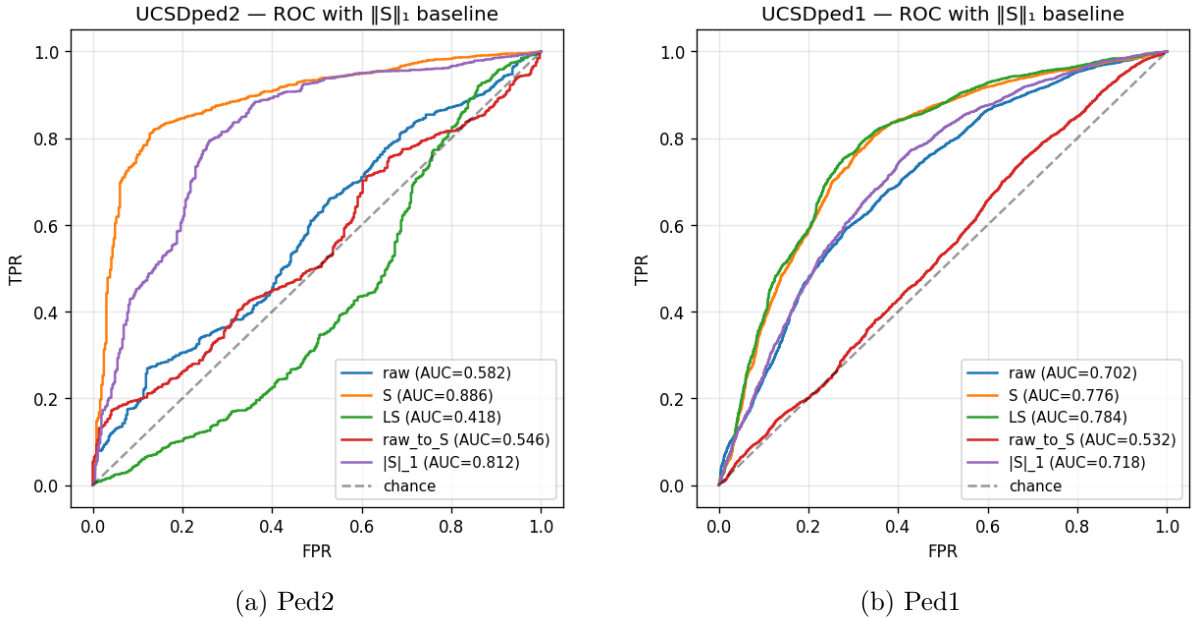


Figure 5: ROC curves for all four learned variants plus the $\|S\|_1$ baseline. The Ped2 panel shows S as the top performer, the $\|S\|_1$ baseline well above raw, and LS below chance. The Ped1 panel shows S and LS tracking closely, raw slightly below, and the $\|S\|_1$ baseline approximately matching raw.

Six observations follow.

(1) RPCA representations beat raw on both datasets. The best RPCA variant exceeds raw by 30 AUC points on Ped2 and 8 on Ped1.

(2) Most of the gain is in the representation, not in the predictor. The non-learned $\|S\|_1$ baseline alone reaches 0.812 on Ped2 and 0.718 on Ped1 — already above raw on both datasets. The best learned variant adds only 7.4 pp (Ped2) and 6.5 pp (Ped1) over this baseline. The headline 30-point Ped2 gain decomposes as roughly +23 pp from the representation and +7 pp from the predictor.

(3) raw_to_S is the worst variant on both datasets. Predicting S from raw input fails: it lands below raw on Ped2 and well below raw on Ped1. The gain from RPCA is realized only when S is on the input side.

(4) The best variant differs by dataset. S is best on Ped2; LS is marginally best on Ped1. We discuss the LS-on-Ped2 collapse in §6.5.

(5) Score distributions confirm the AUC ranking. The histograms in Figure 6 show that the S variant on Ped2 produces visibly separated normal/anomalous distributions (normal concentrated at 0–0.3, anomalous spread 0.3–1.0), and that the $\|S\|_1$ baseline shows a strikingly similar separation pattern — visually consistent with its high AUC. Raw, LS, and raw_to_S show heavily overlapping distributions on Ped2.

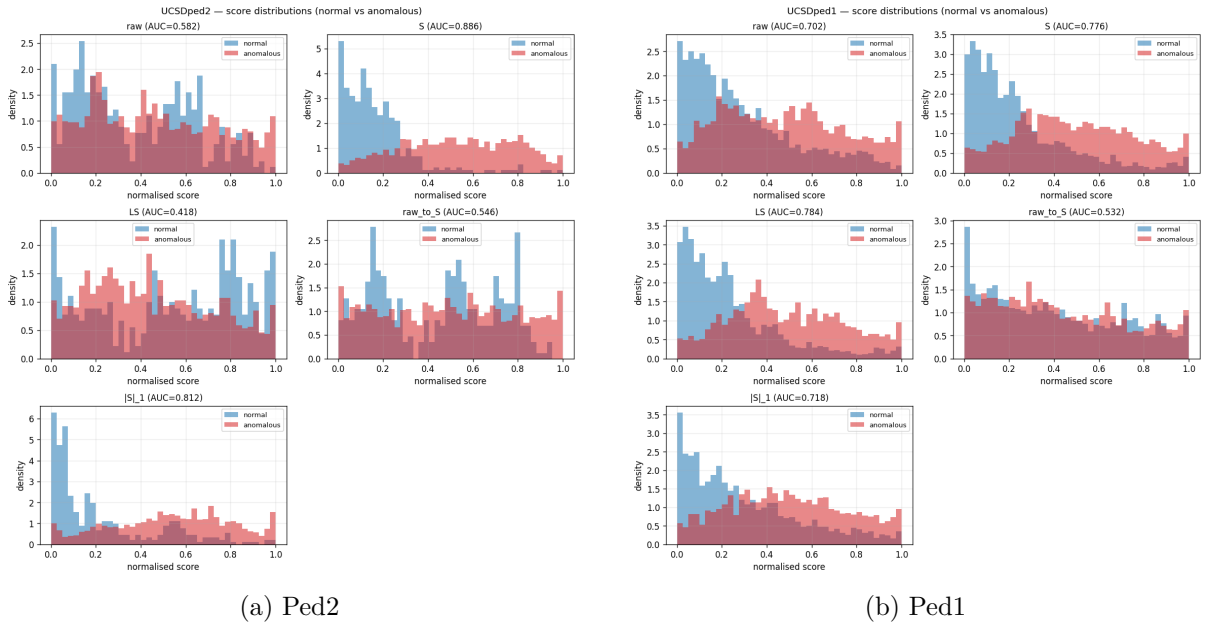
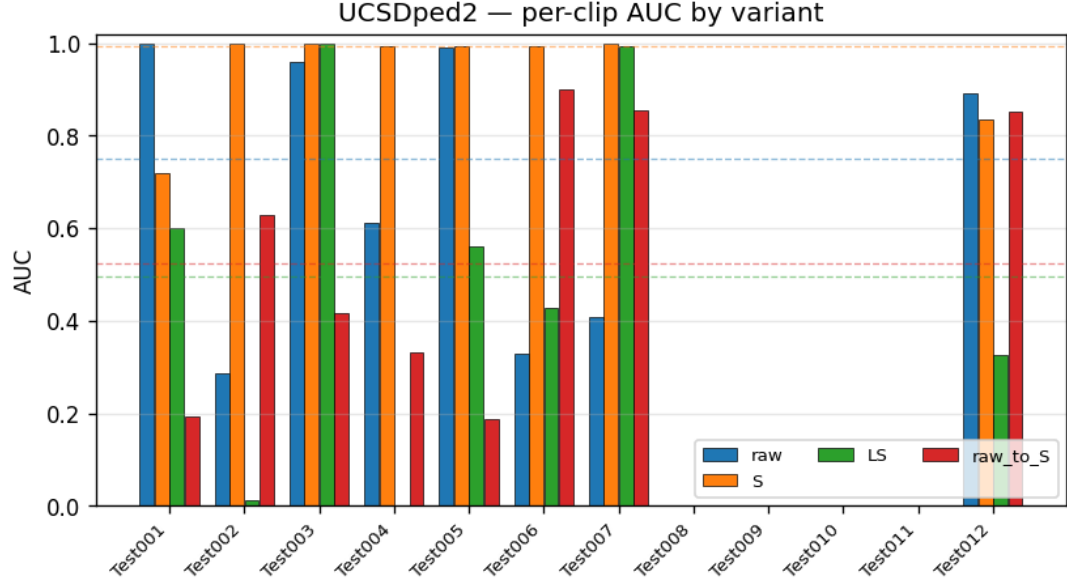
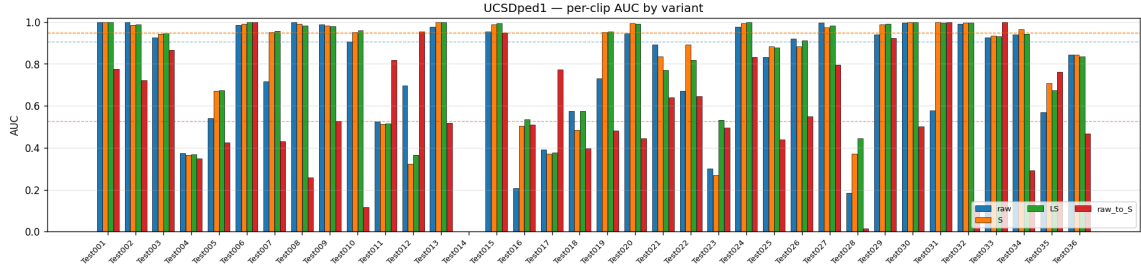


Figure 6: Per-variant histograms of normalized anomaly scores for normal (blue) vs anomalous (red) frames.

(6) Per-clip AUC bars are revealing. On Ped2, the S variant achieves AUC near 1.0 on five of the eight measurable clips. On Ped1, S and LS track each other closely across most clips while raw is markedly worse on a subset.



(a) Ped2



(b) Ped1

Figure 7: Per-clip AUC bar chart for each variant. The Ped2 panel has gaps for Test008–Test011 (100% anomalous, per-clip AUC undefined).

5.3 The Sparse-Component Energy Diagnostic

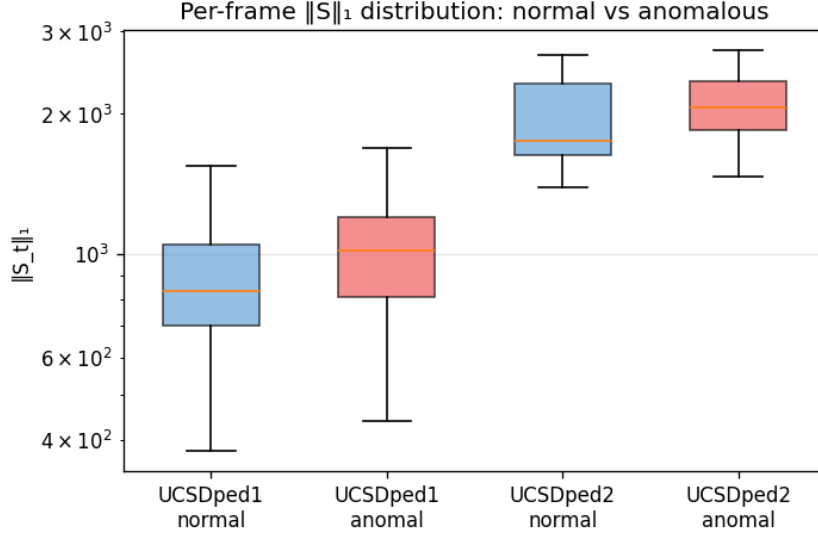


Figure 8: Per-frame $\|S_t\|_1$ boxplots split by ground-truth label, for Ped1 and Ped2 (log y-axis).

On Ped1, anomalous frames have a higher median $\|S\|_1$ than normal frames, consistent with the positive baseline AUC of 0.718. On Ped2, the medians are nearly equal — anomalous boxes sit only marginally higher than normal — which initially looks inconsistent with the 0.812 baseline AUC. The explanation lies in the upper tail: AUC measures rank, and anomalous frames on Ped2 dominate the high end of the $\|S\|_1$ distribution even though typical values overlap. Visually, the anomalous distribution has more mass at high $\|S\|_1$, which is enough to drive ranking-based separability.

5.4 Score Timeline on a Representative Clip

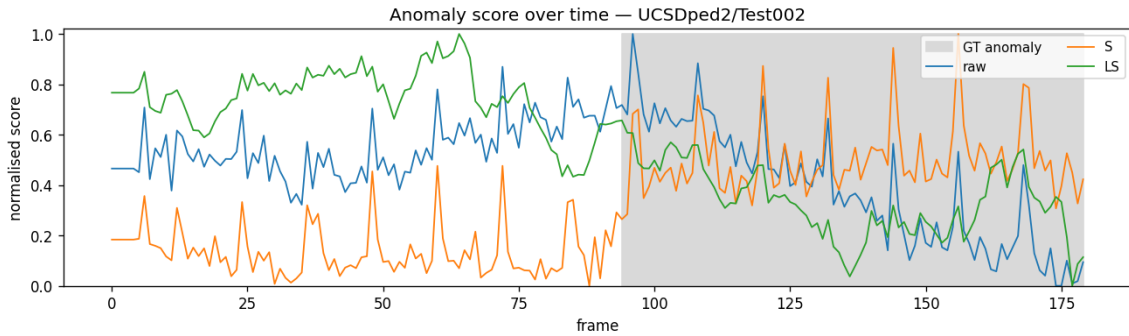


Figure 9: Per-frame anomaly score for the raw, S , and LS predictors on Ped2/Test002, with the GT anomaly window shaded.

Test002 is the test clip with the largest gap between S and raw (per-clip AUC: $S = 1.000$, raw = 0.286). The S model’s score steps up sharply at the anomaly onset and stays elevated; the raw model’s score is dominated by motion-driven noise unrelated to the anomaly. We selected this clip specifically as the largest-gap example; it is illustrative but not representative.

5.5 Failure Case

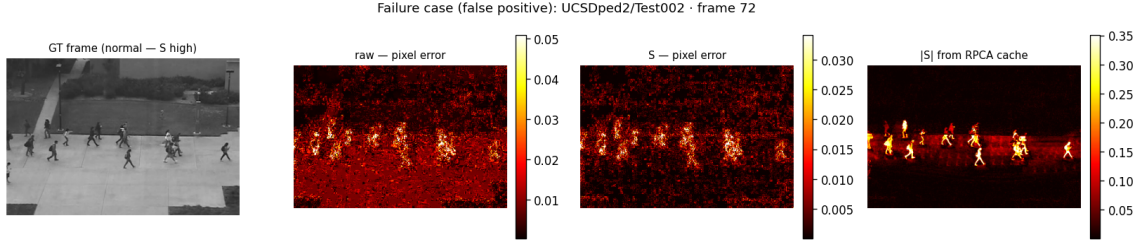


Figure 10: False-positive case on Ped2/Test002 frame 72 (a normal frame). Four panels: GT frame, raw model pixel error, S model pixel error, $|S|$ from the RPCA cache.

This figure shows a normal frame on which the S model erroneously assigns a high score. The pixel-level error map for both models concentrates on pedestrian regions, and the cached $|S|$ panel shows that this frame already has high S energy from a dense pedestrian cluster. This is a direct illustration of the predictor’s confound with the representation: where $\|S\|$ is high for non-anomalous reasons (many pedestrians), the model — having learned a scoring rule that approximates “where is the S energy” — fires.

5.6 SVD Weight Compression (RQ4)

We applied per-layer SVD truncation to every Conv2d weight in the trained Ped2/ S UNet, evaluating frame-level AUC at rank fractions 1.0, 0.75, 0.5, 0.25, 0.1.

Rank fraction	Compression ratio	AUC global	AUC per-clip
1.00	1.115	0.885	0.941
0.75	0.837	0.374	0.328
0.50	0.558	0.697	0.672
0.25	0.280	0.284	0.379
0.10	0.114	0.715	0.690

Table 4: Frame-level AUC versus per-layer SVD truncation rank fraction.

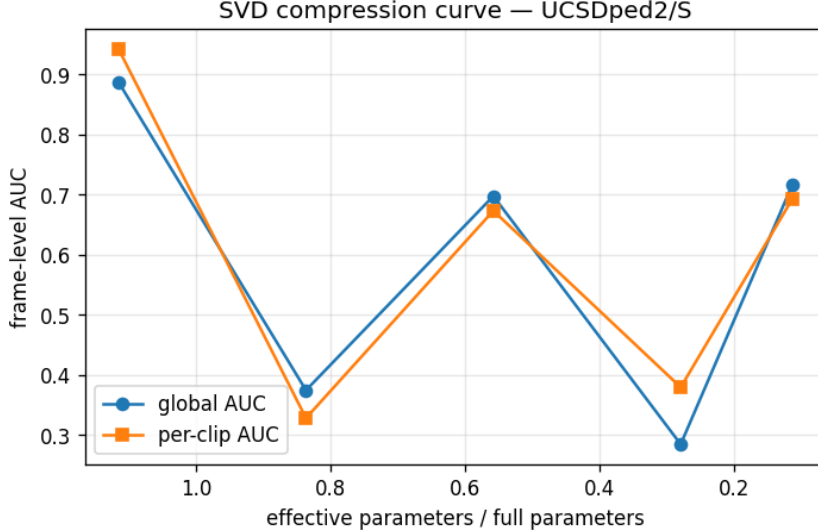


Figure 11: AUC vs effective parameter ratio. The curve is non-monotone: dropping from rank 1.0 to 0.75 collapses AUC, but rank 0.5 partially recovers it, and so on.

We read this curve as a clear **negative result for naïve compression**: the top singular directions of trained Conv2d weights are not aligned with the discriminative directions of the network. Truncation without any compensatory finetuning destroys the model in a non-graceful way. A finetuning step after each truncation would almost certainly produce a smooth degradation curve, but we leave this to future work given the project timeline. The result is reported as-is, and we draw no positive claims about model compression from it.

6 Discussion

6.1 Where the Gain Actually Comes From

The most important finding of this project is that **most of the gain attributable to “RPCA preprocessing” is achievable without any learned model at all**. On Ped2, the simple per-frame anomaly score $\|S_t\|_1$ achieves 0.812 AUC — within 7.4 percentage points of the best learned model and 23 percentage points above the raw-pixel learned baseline. On Ped1, the analogous gap is even tighter: $\|S\|_1$ at 0.718 is 1.6 pp above the raw model and 6.5 pp below the best learned variant.

The implication is not that the learned predictor is useless — its 7-point margin is real and reproducible across both datasets. But it does reframe the contribution of the project: **the dominant effect is that RPCA’s S -component is itself a competent anomaly detector on UCSD**, because the assumptions underlying RPCA (low-rank background, sparse anomalies in the pixel domain) are well-aligned with the structure of the data. The learned predictor refines this signal — it does not create it.

This is consistent with what the score histograms show. The $\|S\|_1$ baseline produces normal/anomalous distributions that closely resemble those of the S -variant predictor. The predictor’s role is to sharpen, not to discover.

6.2 Why the Learned Predictor Helps

If $\|S\|_1$ already separates the classes reasonably, what is the predictor adding? Two effects are plausible.

First, the predictor measures *local* prediction error per frame, not aggregate energy. A frame where pedestrians move predictably will produce low prediction error even if $\|S\|$ is high; a frame where motion is *unpredictable* (such as a cyclist deviating from the pedestrian-flow pattern the model was trained on) will produce high error even if $\|S\|$ is comparable. The predictor’s score therefore captures something orthogonal to raw S energy: predictability of motion in the S domain.

Second, the predictor pools information across the 4-frame window. $\|S\|_1$ is purely instantaneous; the predictor implicitly compares the current S to a temporal context. This helps especially on transitional frames where $\|S\|$ is changing.

The failure case in Figure 10 illustrates the limit of this: when $\|S\|$ is high simply because many pedestrians happen to be in frame, the predictor still fires, because part of its learned scoring rule does track $\|S\|$.

6.3 Why raw_to_S Fails

The `raw_to_S` ablation gives the model raw frames as input and requires it to predict the S component of the next frame. The information needed to compute S is in the input — S is a deterministic function of the clip — but the model would have to re-derive an approximation of RPCA inside its weights while also learning to predict the next frame. This is strictly harder than predicting raw frames (raw target), and the loss penalizes the model for any background it produces while giving it no help in identifying what is background. The ablation thus confirms that the RPCA gain is specifically input-side: the network benefits from the *representation* of S , not from being asked to *produce* S .

6.4 Why Ped2 Gains More Than Ped1

Our project proposal predicted that RPCA would help more on Ped1, where the underlying signal-to-noise ratio is poorer. The opposite is observed: Ped2 sees a 30-point gain (raw $\rightarrow S$) while Ped1 sees only 8 points.

The new $\|S\|_1$ baseline clarifies why. RPCA’s modeling assumptions (low-rank L , sparse S) hold most cleanly when the scene is genuinely sparse — i.e., when the background is mostly empty walkway and only a few moving objects produce S energy. On Ped2 this is the case: anomalies (cyclists, vehicles) produce dramatic $\|S\|$ tail values relative to mostly empty normal frames. On Ped1, normal frames are dense pedestrian crowds, and $\|S\|$ is large even on normal frames; the contrast between normal and anomalous distributions is muted. Both the baseline AUC (Ped2: 0.812 vs. Ped1: 0.718) and the learned-model AUC (Ped2/ S : 0.886 vs. Ped1/ LS : 0.783) reflect this gap.

This reframes RQ3: **the gain from RPCA scales with how cleanly the data matches the low-rank-plus-sparse assumption, not with how noisy the data is in absolute terms.** Ped2 is closer to that ideal than Ped1.

6.5 Why LS Collapses on Ped2

LS provides the predictor with both the static background channel L and the motion channel S , concatenated as a 2-channel input. On Ped2, L is essentially a constant image throughout each clip — visually confirmed in Figure 4, where L is nearly identical across six frames spanning normal and anomalous content. The L channel therefore adds no per-frame predictive information, but does add input dimensionality and presents an attractor for the model to over-fit. On Ped1, LS is competitive with S , plausibly because the larger training set (34 vs. 16 clips) and the richer per-frame content allow the network to make some use of the L channel without overfitting to it. The Ped2 LS result (0.418 — below chance) should be read as a small-data overfitting failure, not as evidence that L hurts in principle.

6.6 Comparison with Public Benchmarks

Reference numbers on UCSD frame-level AUC vary across the literature; the following are widely cited:

- [4], MDT: ~ 0.82 on Ped1, ~ 0.83 on Ped2
- [2], Conv-AE: ~ 0.81 on Ped1, ~ 0.90 on Ped2
- Recent SOTA methods: typically 0.95–0.97 on Ped2

Our best Ped2 result (0.886) is in the range of early deep-learning methods. Our best Ped1 result (0.783) is below the original MDT benchmark. We do not claim state-of-the-art performance, and our methodology — a single UNet predictor without temporal or memory modules — is intentionally simpler than recent SOTA pipelines. The contribution is the controlled study of where the gain originates, supported by the $\|S\|_1$ baseline that is rarely reported in the literature.

It is worth noting that on Ped2 our non-learned $\|S\|_1$ baseline (0.812) is essentially tied with the original MDT benchmark (~ 0.83) — a result obtained with no training and one classical decomposition per clip.

6.7 An Efficiency Interpretation

The decomposition of the Ped2 gain into roughly 23 pp from the representation ($\text{raw} \rightarrow \|S\|_1$) and roughly 7 pp from the learned predictor ($\|S\|_1 \rightarrow S\text{-UNet}$) suggests an interpretation that goes beyond the present experiments. Recent state-of-the-art methods on UCSD reach ~ 0.95 – 0.97 AUC on Ped2, but typically use larger architectures (often 10–50M+ parameters), 3D convolutions or recurrent temporal modules, optical-flow inputs, or memory-bank components. Some of the capacity in these models is plausibly spent on learning an implicit foreground-background decomposition that RPCA provides explicitly and in closed form.

If this is correct, an explicit classical preprocessing step acts as an inductive bias that reduces the effective task complexity for the downstream model: capacity that a raw-input model must allocate to learning what is background can, in an RPCA-preprocessed pipeline, be redirected toward learning what is *unusual* given the foreground signal. The implication would be that a smaller RPCA-preprocessed predictor could reach an AUC comparable to a much larger raw-input model — a parameter-efficiency claim rather than a peak-performance claim. Our 4.5M-parameter UNet reaching 0.886 on Ped2 is consistent with this view but does not establish it. We treat this as an open question for future work and discuss it further in §8.

7 Limitations

- **No temporal model.** The predictor receives a sliding window of 4 frames but has no explicit recurrence. Recent SOTA methods on UCSD use ConvLSTM or transformer-based temporal modelling.
- **No multiple seeds.** Each variant is trained once. Differences of < 2 points between variants should not be considered significant without multi-seed reproduction.
- **Single architecture.** All conclusions are conditioned on a UNet predictor. Whether the input-side / target-side asymmetry would persist with a different architecture is untested.
- **Per-clip AUC on Ped2 excludes 4 clips.** Test008–Test011 contain 100% anomalous frames, making per-clip ROC undefined. The reported per-clip Ped2 AUC is averaged over the remaining 8 clips.
- **SVD compression without finetuning.** The negative compression result is consistent with un-finetuned truncation; a fairer evaluation would include finetuning per truncation level.

8 Future Work

The diagnostic results in §5 motivate a follow-up direction that we did not pursue within the project timeline but consider worth exploring as a hobby project. The central question is whether RPCA preprocessing can substitute for model capacity rather than complement it: that is, whether a small predictor on top of RPCA can match the performance of a much larger raw-input predictor, by virtue of not having to learn an implicit foreground-background decomposition.

A natural protocol is a parameter-matched comparison: train predictors of varying capacity (e.g., 0.1M, 1M, 5M, 20M parameters) with and without RPCA preprocessing on the same data and splits, and report AUC versus parameter count for both pipelines. The hypothesis is that the RPCA-preprocessed curve sits above the raw curve at small parameter counts and that the two curves converge as capacity grows — reflecting that large models can learn an implicit RPCA-equivalent on their own, while small models cannot. If observed, this would frame RPCA preprocessing as a path to similar performance at substantially lower computational cost, a property of practical relevance for deployment on edge hardware where surveillance video anomaly detection is often actually run.

Such a study would also need to test whether the hypothesis holds across architectures (CNN-only, recurrent, transformer) and across datasets where the low-rank-plus-sparse assumption is less clean than on UCSD’s fixed-camera footage.

9 Conclusion

This project demonstrates that the classical matrix-decomposition methods covered in the course — RPCA and SVD — can directly enhance and help interpret a deep-learning anomaly detection pipeline. We applied RPCA per-clip preprocessing to the UCSD Pedestrian Anomaly Detection benchmark and trained a UNet future-frame predictor under four input/target configurations. RPCA preprocessing improves frame-level AUC substantially on Ped2 ($0.582 \rightarrow 0.886$) and modestly on Ped1 ($0.702 \rightarrow 0.783$).

Three diagnostic findings constrain how this gain should be interpreted:

1. A non-learned baseline using only the per-frame ℓ_1 norm of S achieves 0.812 on Ped2 and 0.718 on Ped1 — most of the apparent benefit of “RPCA + UNet” is achievable from the representation alone, with the predictor adding roughly 7 percentage points on top.
2. The `raw_to_S` ablation fails on both datasets, locating the predictor’s contribution specifically on the input side: the network benefits from being given S , not from being asked to produce it.
3. The gain is largest on Ped2, where RPCA’s low-rank-plus-sparse assumption is cleanest. The original proposal hypothesized the opposite; the observation reframes the dependency from “noise level” to “model fit.”

A naïve SVD weight-compression study on the trained predictor produced a non-monotone curve, confirming that finetuning is required after rank truncation if compression is the goal.

Taken together, the results show that a classical, parameter-free decomposition can carry a surprisingly large fraction of the discriminative signal needed for video anomaly detection on a fixed-camera benchmark; that careful baseline construction — separating representation from prediction — is necessary to attribute deep-learning gains correctly; and that the same SVD machinery that drives RPCA also exposes how poorly trained-CNN weights tolerate naïve low-rank truncation. Classical matrix decomposition and modern deep learning sit in productive dialogue throughout the project, with each illuminating something the other obscures.

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